

Year 7 • Chapter 11 • Probability: answers (teacher copy)

11.01 Sample space (booklet p. 3)

WE DO examples (teacher)

- Ex 1a: {H, T}
- Ex 1b: {1, 2, 3, 4, 5, 6}
- Ex 2a: {pass, fail}
- Ex 2b: {G, PG, M, MA15+, R18+}
- Ex 3a: 5; no, the sectors are different sizes
- Ex 3b: {Katrina, Biljana}

Easy • Set A (quick drill)

- | | | |
|--|-------------------------------------|--------|
| 1. {A, B, C, D} | 5. {win, draw, lose} | 8. 6 |
| 2. {Mon, Tue, Wed, Thu, Fri, Sat, Sun} | 6. {summer, autumn, winter, spring} | 9. 12 |
| 3. {a, e, i, o, u} | | 10. 26 |
| 4. {1, 2, 3, 4, 5, 6} | 7. 2 | 11. 5 |
| | | 12. 4 |

Easy • Set B

- 5
- the sectors are different sizes
- 3
- 5
- 1

Medium • Set C

- e.g. {bus, train, ferry}: not equally likely
- {M, A, T, H, S}: equally likely
- {A, P, L, E}: P is more likely
- {0, 1, 2}: 1 is more likely
- {red, amber, green}: not equally likely
- {hearts, diamonds, clubs, spades}: equally likely

Challenging • Set D

- {HH, HT, TH, TT}
- A-H, A-T, B-H, B-T, C-H, C-T, D-H, D-T
- 36 (6×6)
- e.g. sectors of 180° , 90° , 45° , 45°

Skill drill 1 (Easy basics)

- | | | |
|-------|----------------------|-------------------------------------|
| 1. 5 | 10. 10 | 19. 2, 3, 5, 7, 11 |
| 2. 5 | 11. 4 | 20. 1, 3, 5, 7, 9 |
| 3. 52 | 12. 7 | 21. hearts, diamonds, clubs, spades |
| 4. 26 | 13. 5 | |
| 5. 7 | 14. 6 | |
| 6. 8 | 15. 5 | 22. H, T |
| 7. 4 | 16. B, O, K | 23. C, A, T |
| 8. 12 | 17. red, blue, green | 24. Saturday, Sunday |
| 9. 2 | 18. 1, 2, 3, 4, 5, 6 | 25. a, e, i, o, u |

Skill drill 2 (Medium basics)

- | | | |
|-------|---------|-----------------|
| 1. 4 | 9. 30 | 17. no |
| 2. 84 | 10. 12 | 18. no |
| 3. 18 | 11. 12 | 19. yes (about) |
| 4. 14 | 12. 36 | 20. yes |
| 5. 8 | 13. no | 21. yes |
| 6. 8 | 14. yes | 22. no |
| 7. 8 | 15. no | |
| 8. 16 | 16. yes | |

Extension

- 16
- 9

Exit ticket

- Level 1: {red, yellow, blue}
- Level 2: 6
- Level 3: no: it depends on the student

11.02 Probability (booklet p. 9)

WE DO examples (teacher)

- Ex 1a: $\frac{1}{6}$
- Ex 1b: $\frac{1}{2}$
- Ex 2a: $\frac{2}{3}$
- Ex 2b: $\frac{3}{10}$ (C)
- Ex 3a: 20.2%
- Ex 3b: 86.1%

Easy • Set A (quick drill)

- | | | |
|------------------|-------------------|-------------------|
| 1. $\frac{1}{6}$ | 5. 1 | 9. $\frac{1}{5}$ |
| 2. $\frac{1}{2}$ | 6. $\frac{1}{3}$ | 10. $\frac{1}{2}$ |
| 3. $\frac{2}{3}$ | 7. $\frac{3}{10}$ | 11. 0 |
| 4. 0 | 8. $\frac{1}{2}$ | 12. $\frac{1}{2}$ |

Easy • Set B

- $\frac{2}{11}$
- $\frac{2}{11}$
- $\frac{4}{11}$
- $\frac{1}{11}$
- $\frac{4}{11}$
- 0

Medium • Set C

- $\frac{3}{10}$, 30.0%
- $\frac{3}{8}$, 37.5%
- $\frac{27}{40}$, 67.5%
- $\frac{21}{104}$, 20.2%
- $\frac{137}{208}$, 65.9%
- 0, 0%

Challenging • Set D

- 0.44
- 1.6%
- 0.65
- 3

Skill drill 1 (Easy basics)

- | | | |
|-------------------|--------------------|--------------------|
| 1. $\frac{7}{10}$ | 9. $\frac{4}{5}$ | 17. $\frac{1}{5}$ |
| 2. $\frac{3}{5}$ | 10. $\frac{3}{7}$ | 18. $\frac{1}{2}$ |
| 3. $\frac{8}{9}$ | 11. $\frac{1}{3}$ | 19. $\frac{3}{4}$ |
| 4. $\frac{2}{3}$ | 12. $\frac{1}{6}$ | 20. $\frac{9}{13}$ |
| 5. $\frac{1}{3}$ | 13. $\frac{8}{17}$ | 21. $\frac{4}{9}$ |
| 6. $\frac{1}{4}$ | 14. $\frac{4}{7}$ | 22. $\frac{1}{3}$ |
| 7. $\frac{2}{3}$ | 15. $\frac{3}{4}$ | 23. $\frac{9}{13}$ |
| 8. $\frac{1}{5}$ | 16. $\frac{1}{5}$ | 24. $\frac{7}{12}$ |

25. $\frac{1}{3}$
 26. $\frac{1}{3}$
 27. 0
 28. $\frac{1}{3}$
 29. $\frac{1}{2}$
 30. $\frac{1}{2}$

31. $\frac{1}{6}$
 32. 1
 33. $\frac{1}{3}$
 34. $\frac{2}{3}$
 35. $\frac{1}{3}$
 36. 1

37. $\frac{1}{2}$
 38. $\frac{5}{6}$
 39. $\frac{1}{6}$
 40. $\frac{1}{3}$

Skill drill 2 (Medium basics)

1. $\frac{1}{9}$	10. $\frac{1}{11}$	19. $\frac{19}{20} = 0.95 = 95\%$
2. $\frac{1}{4}$	11. $\frac{1}{10}$	20. $\frac{9}{10} = 0.9 = 90\%$
3. $\frac{1}{2}$	12. $\frac{1}{5}$	21. $\frac{3}{25} = 0.12 = 12\%$
4. $\frac{1}{10}$	13. $\frac{1}{9}$	22. $\frac{1}{4} = 0.25 = 25\%$
5. $\frac{1}{8}$	14. $\frac{1}{9}$	23. $\frac{7}{10} = 0.7 = 70\%$
6. $\frac{1}{11}$	15. $\frac{2}{11}$	24. $\frac{3}{4} = 0.75 = 75\%$
7. $\frac{1}{10}$	16. $\frac{4}{5} = 0.8 = 80\%$	25. $\frac{23}{25} = 0.92 = 92\%$
8. $\frac{1}{10}$	17. $\frac{8}{25} = 0.32 = 32\%$	26. $\frac{3}{20} = 0.15 = 15\%$
9. $\frac{3}{8}$	18. $\frac{1}{2} = 0.5 = 50\%$	27. $\frac{9}{25} = 0.36 = 36\%$

Extension

1. $\frac{4}{13}$
 2. $\frac{3}{13}$

Exit ticket

- Level 1: $\frac{1}{6}$
- Level 2: $\frac{3}{10}$
- Level 3: 30%

11.03 The probability scale (booklet p. 15)

WE DO examples (teacher)

- Ex 1a: B
- Ex 1b: unlikely
- Ex 2a: rolling an odd or even number on a die (C)
- Ex 2b: $\frac{3}{7}$
- Ex 3a: $\frac{3}{14}$
- Ex 3b: 0; 1

Easy · Set A (quick drill)

- certain
- impossible
- even chance
- very unlikely
- likely
- certain

Easy · Set B

- impossible
- unlikely
- even chance
- likely
- certain
- unlikely

Medium · Set C

1. $\frac{3}{7}$
 2. $\frac{3}{14}$

3. 0
 4. 1
 5. $\frac{5}{14}$
 6. $\frac{4}{7}$

Challenging · Set D

- A at $\frac{1}{2}$, B at $\frac{1}{6}$, C at 1, D at 0
- own examples

Skill drill 1 (Easy basics)

- | | | |
|---------------|-----------------|-----------------------------------|
| 1. likely | 13. likely | 25. even chance |
| 2. likely | 14. unlikely | 26. unlikely |
| 3. unlikely | 15. even chance | 27. likely |
| 4. impossible | 16. even chance | 28. unlikely |
| 5. certain | 17. unlikely | 29. even chance |
| 6. unlikely | 18. unlikely | 30. impossible |
| 7. likely | 19. likely | 31. likely |
| 8. unlikely | 20. likely | 32. even chance |
| 9. likely | 21. unlikely | 33. certain |
| 10. likely | 22. likely | 34. certain |
| 11. likely | 23. unlikely | 35. impossible (or very unlikely) |
| 12. likely | 24. certain | 36. unlikely |

Skill drill 2 (Medium basics)

- | | | |
|------------------|--------------------|-----------------|
| 1. $\frac{5}{8}$ | 9. $\frac{3}{5}$ | 18. unlikely |
| 2. $\frac{2}{5}$ | 10. $\frac{7}{10}$ | 19. unlikely |
| 3. 45% | 11. 65% | 20. likely |
| 4. $\frac{3}{8}$ | 12. 80% | 21. unlikely |
| 5. 45% | 13. $\frac{5}{8}$ | 22. likely |
| 6. $\frac{2}{5}$ | 14. $\frac{1}{4}$ | 23. unlikely |
| 7. $\frac{5}{8}$ | 15. $\frac{1}{3}$ | 24. certain |
| 8. 40% | 16. 90% | 25. likely |
| | 17. certain | 26. likely |
| | | 27. unlikely |
| | | 28. even chance |

Extension

- 4
- e.g. 2 blue, 7 red, 1 green

Exit ticket

- Level 1: 0
- Level 2: very likely
- Level 3: $\frac{5}{8}$

11.04 Experimental probability (booklet p. 21)

WE DO examples (teacher)

- Ex 1a: $\frac{1}{2}$
- Ex 1b: 40
- Ex 2a: $\frac{3}{10}$
- Ex 2b: fewer heads than expected (24 vs 40)
- Ex 3a: 10
- Ex 3b: the results even out over many trials

Easy · Set A (quick drill)

- | | | |
|------------------|------------------|---------|
| 1. $\frac{1}{2}$ | 4. $\frac{1}{2}$ | 8. 5 |
| 2. $\frac{1}{6}$ | 5. $\frac{3}{5}$ | 9. 20 |
| 3. $\frac{1}{4}$ | 6. $\frac{2}{5}$ | 10. 25 |
| | 7. 25 | 11. 100 |
| | | 12. 20 |

Easy · Set B

- 40

2. 10
3. 25
4. 50
5. 60
6. 500

Medium · Set C

1. $\frac{3}{10}$
2. $\frac{1}{4}$
3. $\frac{21}{50}$
4. $\frac{7}{20}$
5. $\frac{13}{40}$
6. 0

Challenging · Set D

1. own results
2. answers vary; class results close to $\frac{1}{2}$

Skill drill 1 (Easy basics)

- | | | |
|--------|----------------------|---------------------|
| 1. 23 | 18. 17 | 31. $\frac{13}{20}$ |
| 2. 3 | 19. 27 | 32. $\frac{37}{50}$ |
| 3. 18 | 20. 9 | 33. $\frac{3}{4}$ |
| 4. 4 | 21. 26 | 34. $\frac{7}{50}$ |
| 5. 22 | 22. 12 | 35. $\frac{7}{20}$ |
| 6. 18 | 23. 13 | 36. $\frac{9}{20}$ |
| 7. 26 | 24. 18 | 37. $\frac{23}{60}$ |
| 8. 4 | 25. $\frac{11}{40}$ | 38. $\frac{4}{5}$ |
| 9. 9 | 26. $\frac{19}{30}$ | 39. $\frac{8}{15}$ |
| 10. 10 | 27. $\frac{41}{60}$ | 40. $\frac{2}{3}$ |
| 11. 29 | 28. $\frac{39}{100}$ | |
| 12. 21 | 29. $\frac{7}{30}$ | |
| 13. 23 | 30. $\frac{9}{20}$ | |
| 14. 18 | | |
| 15. 27 | | |
| 16. 24 | | |
| 17. 15 | | |

Skill drill 2 (Medium basics)

- | | | |
|----------------------------|-----------------------------|---------|
| 1. 26% (theory
16.7%) | 7. 16% (theory
16.7%) | 13. 20 |
| 2. 17% (theory
16.7%) | 8. 21% (theory
16.7%) | 14. 16 |
| 3. 22% (theory
16.7%) | 9. 18% (theory
16.7%) | 15. 11 |
| 4. 40% (theory
16.7%) | 10. 10% (theory
16.7%) | 16. 27 |
| 5. 30% (theory
16.7%) | 11. 12.5% (theory
16.7%) | 17. 70 |
| 6. 17.5% (theory
16.7%) | 12. 35% (theory
16.7%) | 18. 18 |
| | | 19. 57 |
| | | 20. 105 |
| | | 21. 16 |
| | | 22. 16 |
| | | 23. 7 |
| | | 24. 35 |

Extension

1. about 150
2. no: red has $P \approx 0.3$, not 0.25

Exit ticket

1. Level 1: $\frac{1}{6}$
2. Level 2: 20
3. Level 3: $\frac{5}{24}$

11.05 Complementary events (booklet p. 27)

WE DO examples (teacher)

1. Ex 1a: not a queen, $\frac{12}{13}$

2. Ex 1b: not a multiple of 3, $\frac{2}{3}$

3. Ex 2a: not winning, $\frac{549}{550}$

4. Ex 2b: $\frac{1}{5}$

5. Ex 3a: $\frac{3}{5}$; $\frac{3}{5}$

6. Ex 3b: $\frac{7}{15}$

Easy · Set A (quick drill)

- | | | |
|-------------------------|--------------------------|-------------------|
| 1. not rolling a 6 | 5. choosing a consonant | 8. $\frac{4}{5}$ |
| 2. getting a tail | 6. rolling an odd number | 9. 55% |
| 3. drawing a black card | 7. 0.7 | 10. 1 |
| 4. losing | | 11. $\frac{5}{8}$ |
| | | 12. 0.01 |

Easy · Set B

1. $\frac{3}{4}$
2. 0.7
3. 55%
4. $\frac{7}{12}$
5. 0.93
6. 0

Medium · Set C

1. $\frac{12}{13}$
2. $\frac{2}{3}$
3. $\frac{549}{550}$
4. $\frac{4}{5}$
5. $\frac{7}{15}$
6. $\frac{3}{4}$

Challenging · Set D

1. 0.65
2. 0.3
3. 8
4. an event either happens or it does not

Skill drill 1 (Easy basics)

- | | | |
|---------------------|---------------------|----------|
| 1. $\frac{4}{5}$ | 13. $\frac{5}{8}$ | 25. 0.46 |
| 2. $\frac{3}{4}$ | 14. $\frac{1}{15}$ | 26. 0.56 |
| 3. $\frac{2}{7}$ | 15. $\frac{3}{5}$ | 27. 62% |
| 4. $\frac{5}{6}$ | 16. $\frac{1}{12}$ | 28. 80% |
| 5. $\frac{1}{4}$ | 17. $\frac{13}{15}$ | 29. 44% |
| 6. $\frac{1}{6}$ | 18. $\frac{5}{12}$ | 30. 0.1 |
| 7. $\frac{1}{8}$ | 19. $\frac{1}{3}$ | 31. 0.98 |
| 8. $\frac{11}{12}$ | 20. $\frac{3}{8}$ | 32. 55% |
| 9. $\frac{1}{2}$ | 21. $\frac{7}{8}$ | 33. 36% |
| 10. $\frac{4}{7}$ | 22. $\frac{1}{20}$ | 34. 0.01 |
| 11. $\frac{19}{20}$ | 23. $\frac{7}{12}$ | 35. 57% |
| 12. $\frac{6}{7}$ | 24. $\frac{1}{5}$ | 36. 78% |
| | | 37. 0.49 |
| | | 38. 0.44 |
| | | 39. 0.37 |
| | | 40. 41% |

Skill drill 2 (Medium basics)

- | | | |
|-------------------------|------------------------------------|---------------------|
| 1. choosing a consonant | 2. not winning (losing or drawing) | 3. it does not rain |
|-------------------------|------------------------------------|---------------------|

- | | | |
|-------------------------------------|---------------------------------------|---------------------|
| 4. choosing a black card | 10. choosing a club, diamond or spade | 16. $\frac{8}{11}$ |
| 5. not rolling a 6 (rolling 1 to 5) | | 17. $\frac{3}{7}$ |
| 6. tossing tails | 11. $\frac{7}{10}$ | 18. $\frac{11}{14}$ |
| 7. failing the test | 12. $\frac{10}{19}$ | 19. $\frac{9}{11}$ |
| 8. rolling 3 or more | 13. $\frac{8}{13}$ | 20. $\frac{7}{8}$ |
| 9. rolling an odd number | 14. $\frac{8}{17}$ | 21. $\frac{4}{7}$ |
| | 15. $\frac{4}{7}$ | 22. $\frac{11}{17}$ |

Extension

1. $\frac{5}{12}$
2. 130

Exit ticket

1. Level 1: not rolling a 2
2. Level 2: 0.2
3. Level 3: $\frac{1}{2}$